

Effects of Ability and Family Circumstances on Earnings Potential

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I Introduction

This paper looks to address (i) how strongly one's future earnings potential is affected by ability, and (ii) how strongly one's future earnings are affected by having a single mother. Ability is an unobservable, complex concept, and is impossible to measure as a variable. For our purposes, we will use the IQ and Knowledge of World of Work (KWW) scores as a proxy measure of ability. We also observe if a child had a single mother at the age of fourteen (sinmom14 variable) to proxy for a child's upbringing. We expect KWW, IQ, and sinmom14 to hold great weight in determining future earnings potential. We expect the controls education and experience to have significant effects as well. The questions at hand are important as their answers can aid in the development of public policies aiming to better the lives of lower-income Americans. Such policies could be focused on bolstering nutrition, education, career opportunities, or one's ability to move upward in society.

The underlying theoretical model we use relies on assumptions generally made by society at large. Those being that there is a positive correlation between IQ and KWW scores with earnings, and a negative correlation between having a single mother and earnings. In order to test these hypotheses, we will run regressions in SAS that try to find the effect on log of wage (lwage) depending on KWW and IQ score, and the given single mother variable. For controls in this experiment we considered race, region, education level, experience, and marital status.

II Data

David Card got the data from the National Longitudinal Survey of Young Men (NLSYM). Beginning in 1966, researchers followed the subjects through 1981 to track how their lives changed over that time. It should be noted that the NLSYM is not a random sample, and therefore it is not perfectly representative of the U.S. population. The survey over-sampled neighborhoods with a high concentration of non-white residents (Card, 1993). Our

data suffers from sample attrition; within the first three years of the survey's start, twenty percent of the sample had dropped out (Card, 1993). By the 1981 conclusion, only sixty-five percent of the sample was interviewed (Card, 1993). Due to missing values of both IQ and KWW, we had to restrict the data set to 2,040 observations out of 3,010. 83.2 percent of the sample lived with both parents at age fourteen, while only eight percent lived with a single mom at age fourteen. The average IQ score of our sample is 102.4 and the average KWW score of our sample is 35.6. The average person in our sample is twenty-eight years old and has 13.92 years of education, suggesting the average person had at least some college education. The maximum education was eighteen years, suggesting that at least one person went to graduate school. 14 percent of our sample was African American, and Region 663 was home to the majority of our sample, at 21.9 percent. 34 percent of our sample lived in the southern half of the U.S, and 74 percent of our sample lived in a standard metropolitan statistical area (SMSA). Experience is measured as age - educ - 6, with the average being 8.5. Married is not a binary variable, but rather it has six possible values. Interestingly, the average hourly wage was 614.2 cents, with the minimum being 112 and the maximum being 2,404.

III The Model

Our econometric model used log wage as the dependent variable. Lwage is equal to the log of the hourly wages in cents of the sample. We chose to use the log of wage as it simplifies the output model by promoting linear relationships. We regressed this on black, smsa, south, educ, exper, exper2, married, KWW, IQ, momdad14, sinmom14, reg661, reg662, reg663, reg664, reg665, reg666, reg667, and reg669. We chose KWW and IQ as they are proxies for ability, an extremely important variable we cannot measure. We added sinmom14 to measure the effect of a single mother. We chose black, smsa, south, educ, exper, married, momdad14, reg661-669 as we believe them to be the best controls for our model. Note that

region 668 was not included, we omitted it as it makes our model not full rank; it took on the value of the intercept. Finally, we included `exper2` as a correction for non-linearity.

We used Ordinary Least Squares estimators throughout the analysis. A semi-log model is used in all models but one, with the dependent variable being log wage. We tested for nonlinearity, and use power terms to address it. We used the White Test to check for heteroskedasticity, and attempted to find instrumental variables using a two stage least regression. Our model identifies the answer using relevant variables (`KWW`, `IQ`, and `sinmom14`) and controlled to try and estimate the effects on log wage. Our first regression was our base model. In this model, regressed log wage on `black`, `smsa`, `south`, `educ`, `exper`, `married`, `reg661`, `reg662`, `reg663`, `reg664`, `reg665`, `reg666`, `reg667`, `reg669`. As noted above, region 668 was omitted and took the value of the intercept. We used an F-test to see if our regions were jointly significant. With an f-value of 3.72 and a probability value of 0.0005, we found that all regions are jointly significant. Turning to the rest of the regression, the majority of the variables were statistically significant. Some of the parameters for the variables were negative; they are `black`, `south` and `married`. This means there is a negative premium on log wage for each of those variables. Besides that, the intercept had a very high t-value (52.04) and parameter estimate as compared to all other variables, meaning a lot of explanatory power remains to be attributed to the variables not included in this regression.

In the second regression, we checked for non-linearity with the `exper2` variable created in the `carddata` dataset. It is equal to experience squared. Since the absolute value of t for `exper2` is greater than 2 (4.70), `exper2` is a significant variable. This suggested that we have detected non linearity in our model. Additionally, the F-test for `exper` and `exper2` coefficients to be zero gave a probability value of <0.0001 , with an f-value of 109.92. This proved that education is a significant determinant of log wage. However, we chose to only include up to `exper2` since we believed that it solved the issue of non-linearity in our semi-log model.

Moving to our fifth regression, we added in both `KWW` and `IQ`. We added the variables

separately and found that they were significant on their own in the third and fourth regression. Here, upon adding KWW alongside IQ, the t value of IQ decreased from 3.71 to 2.44, and KWW decreased from 5.31 to 4.47. This suggested that there was explanatory power incorrectly attributed to both variables that was being explained by the other. To further test the significance of our proxies for ability, we ran an F test on KWW and IQ. The F statistic value was 17.10 and the probability value was < 0.0001 . This implied that KWW and IQ are jointly significant, and were significant determinants of log wage. We then used a White Test to check for heteroskedasticity and found that, with a probability value of 0.2481, we failed to reject the null at a significance level of 5 percent and found that there is no heteroskedasticity or nonlinearity. For KWW and IQ, the parameter estimates are positive, which is in-line with our expectations. For KWW, the estimate is 0.00627 and for IQ the estimate is 0.00169. We believe that we have economic and statistical significance for KWW and IQ. The education parameter estimate is positive at 0.05774, which matches our expectations. The black variable had a negative parameter estimate at -0.10666. This suggested that being African American has a negative effect on future earnings potential. Marriage is very significant with a t-value of -7.05, and its coefficient is negative at -0.02949, which also suggests that getting married can negatively affect future earnings potential. Interestingly, experience is also very significant at a t-value of 7.47 and has a positive coefficient at 0.07306. However, experience squared has a negative coefficient at -0.00212. When calculating AIC, our model has a value of -4143.1017, which is the second lowest out of all possible combinations. We believe this to be evidence that we have not overfit the data. When compared to the AIC of the models with only IQ and only KWW, the AIC in our preferred model is the lowest. While the differences were not very large between the three models, we chose to include both KWW and IQ as we believe that they are the best measures to correct the problem of the unobserved variable of ability. Furthermore, the adjusted R square for our preferred model is the highest at 0.2555. The t-values for education, experience, exper2,

and the intercept all decrease compared to our base model, suggesting that KWW and IQ explain more of the variation on log wage.

Next, in the sixth regression, we added in the single mother variable (sinmom14) on our model from regression 5. Notably it was insignificant, with a t-value of -0.83. Its probability value is 0.4088, so we strongly failed to reject the null hypothesis. In the seventh regression, we put in the mom and dad variable only as we believe that it is a necessary control for single mom. It was insignificant at a t-value of 1.78, and had a probability value of 0.0750.

In our eighth regression, we included both the single mother and mom and dad variables. Our key finding here was that both the parent variables were insignificant even when brought together. Also, when performing an F test, we found that they are jointly insignificant. We then used a White test to check for heteroskedasticity (and by extension non-linearity), and found that we had neither at a 5 percent significance level as our probability value was 0.3613. Our parameter estimates for KWW and IQ were 0.00623 and 0.00169 respectively. This is in-line with our expectations and the findings in our model that addressed ability. Black and married remained negative, while experience remained positive. All significant variables in our first preferred model remained significant here. Education's parameter estimate is 0.05708, and remained positive. We ultimately chose to include both parent variables because we believe that they give the best picture of what we are trying to answer, as it has one of the highest adjusted r square values of the single mom regressions at 0.2560. The findings of this model barely differed from the findings from our first preferred model. In the ninth and final regression, we attempted to identify instrumental variables. We suspected that the nearc2 and nearc4 variables were instrumental; nearc2 is defined as living near a two year college, and nearc4 is defined as living near a four year college. We also suspected that education is endogenous, and to test this we used a two stage least squares test. In stage one, we regressed education on our variables from our preferred single mother model (eighth regression) and the nearc2 and nearc4 variables. We found that nearc2 is insignificant with a t-value of

0.68, and a probability value of 0.4969. Nearc4 was significant with a t-value of 3.28, and a probability value of 0.0011. This implied that nearc2 is not a valid instrument. To further test our findings, we used an F test to see if the two variables were jointly significant. The f-value was 5.88, which is less than the desired f-value of 10. Therefore, we had reason to believe that nearc2 and nearc4 were very weak, and may not be valid instrumental variables. To further confirm this, we moved on to stage two of the two stage least squares test. After running the test, our standard error on education went from 0.00539 in our preferred single mother model to 0.074460 in the two-stage least squares test model. Our standard error on IQ went from 0.00069466 to 0.001967. Due to the result of the f-test being half of what we desired, and our education standard error increasing by thirteen-fold, we found that nearc2 and nearc4 are invalid instruments.

Our final preferred model is our eighth regression. In this model we regressed log wage on black, smsa, south, educ, exper, exper2, married, KWW, IQ, momdad14, sinmom14, reg661, reg662, reg663, reg664, reg665, reg666, reg667, and reg669. We chose this model to be our preferred model since it had all variables and controls we needed to answer our hypotheses. It also had the higher adjusted r square value versus our KWW and IQ model.

IV Discussion

Our original hypothesis was found to be partially correct, as there is a positive correlation between log wage with IQ and KWW. However, there is no statistical significance of having a single mother at age fourteen. That led us to reject the notion that having a single mother hinders one's future earnings potential. However, that conflicts with the commonly held belief suggesting that there is such a correlation. This naturally brings up a few questions about the sinmom14 variable. It is possible that there is truly little to no correlation between one's earnings and whether or not they had a single mother. It is also possible that this specific variable is flawed (maybe a child had both parents for most of their upbringing

before turning 14), or that the methods used in recording the data were flawed. It should be noted that the adjusted R-square value of our model indicates that roughly three quarters of log wage is not derived from the model. There is a high level of randomness at play, and our variables are more suggestive and not determinative. We also reiterate that there was an attrition bias in the original survey, and that the survey wasn't a random sample of the United States population. Furthermore, we restricted the data down to 2,040 observations, when the original had 3,010. There is more research needed before a definitive conclusion on the subject could be made. Given what we found here, it makes sense to focus on the KWW and IQ portion of our hypothesis. Economists and politicians could point to our findings when discussing how to better the economic wellbeing of our nation's people. Educators could also point to our findings, and better their efforts to match students to appropriate and fitting career paths.

Table 1: Proc Means Table

Variable	N	Mean	Std Dev	Minimum	Maximum
id	2040	2391.1600000	1468.6200000	3.0000000	5221.0000000
nearc2	2040	0.4539216	0.4979943	0	1.0000000
nearc4	2040	0.7078431	0.4548655	0	1.0000000
educ	2040	13.9240196	2.2771681	8.0000000	18.0000000
age	2040	28.3852941	2.9970689	24.0000000	34.0000000
fatheduc	1654	10.5816203	3.3866112	0	18.0000000
motheduc	1860	10.8741935	2.8842248	0	18.0000000
weight	2040	346362.0600000	161994.6900000	75607.0000000	1693822.0000000
momdad14	2040	0.8323529	0.3736442	0	1.0000000
sinmom14	2040	0.0803922	0.2719660	0	1.0000000
step14	2040	0.0372549	0.1894322	0	1.0000000
reg661	2040	0.0509804	0.2200116	0	1.0000000
reg662	2040	0.1877451	0.3906042	0	1.0000000
reg663	2040	0.2191176	0.4137499	0	1.0000000
reg664	2040	0.0740196	0.2618670	0	1.0000000
reg665	2040	0.1779412	0.3825570	0	1.0000000
reg666	2040	0.0647059	0.2460665	0	1.0000000
reg667	2040	0.0926471	0.2900083	0	1.0000000
reg668	2040	0.0328431	0.1782696	0	1.0000000
reg669	2040	0.1000000	0.3000736	0	1.0000000
south66	2040	0.3352941	0.4722089	0	1.0000000
black	2040	0.1436275	0.3507976	0	1.0000000
smsa	2040	0.7401961	0.4386344	0	1.0000000
south	2040	0.3411765	0.4742208	0	1.0000000
smsa66	2040	0.6906863	0.4623240	0	1.0000000
wage	2040	614.2009804	263.8042336	112.0000000	2404.0000000
enroll	2040	0.1053922	0.3071333	0	1.0000000
KWW	2040	35.6019608	7.5396221	10.0000000	56.0000000
IQ	2040	102.4455882	15.4323336	50.0000000	149.0000000
married	2039	2.1971555	2.0176775	1.0000000	6.0000000
libcrd14	2034	0.7586037	0.4280353	0	1.0000000
exper	2040	8.4612745	3.7373911	0	19.0000000
lwage	2040	6.3347600	0.4177649	4.7184990	7.7848890
expersq	2040	85.5544118	69.8887996	0	361.0000000
exper2	2040	85.5544118	69.8887996	0	361.0000000

Table 2: Regression Output

	(1) OLS	(2) OLS	(5) OLS	(6) OLS	(7) OLS	(8) OLS
Black Student (BLACK)	-0.16446* (0.02489)	-0.16672* (0.02476)	-0.10666* (0.02693)	-0.10292* (0.02731)	-0.09663* (0.02749)	-0.09713* (0.02752)
SMSA	0.15766* (0.01928)	0.15748* (0.01918)	0.14729* (0.01911)	0.14767* (0.01912)	0.14725* (0.01910)	0.14699* (0.01912)
SOUTH	-0.11027* (0.03182)	-0.10761* (0.03166)	-0.10693* (0.03142)	-0.10732* (0.03142)	-0.10883* (0.03142)	-0.10895* (0.03142)
Education (EDUC)	0.07239* (0.00451)	0.07508* (0.00452)	0.05774* (0.00538)	0.05755* (0.00539)	0.05709* (0.00539)	0.05708* (0.00539)
Experience (EXPER)	0.03879* (0.00277)	0.08164* (0.00953)	0.07306* (0.00978)	0.07323* (0.00978)	0.07330* (0.00977)	0.07323* (0.00977)
Experience ² (EXPER2)		-0.00229* (0.00048837)	-0.00212* (0.00048784)	-0.00212* (0.00048797)	-0.00213* (0.00048762)	-0.00212* (0.00048777)
MARRIED	-0.03157* (0.00417)	-0.02974* (0.00417)	-0.02949* (0.00413)	-0.02951* (0.00413)	-0.02923* (0.00413)	-0.02917* (0.00414)
KWW			0.00627* (0.00140)	0.00629* (0.00140)	0.00625* (0.00140)	0.00623* (0.00140)
IQ			0.00169* (0.00069453)	0.00171* (0.00069483)	0.00170* (0.00069416)	0.00169* (0.00069466)
MOMDAD14					0.03965 (0.02226)	0.04758 (0.02913)
SINMOM14				-0.02478 (0.03000)		0.01655 (0.03924)
REG661	0.06332 (0.05731)	0.07133 (0.05704)	0.05130 (0.05672)	0.05227 (0.05674)	0.05130 (0.05669)	0.05065 (0.05673)
REG662	0.16035* (0.04855)	0.16383* (0.04830)	0.15432* (0.04805)	0.15461* (0.04806)	0.15227* (0.04804)	0.15167* (0.04807)
REG663	0.17885* (0.04803)	0.18272* (0.04779)	0.17412* (0.04750)	0.17421* (0.04750)	0.17277* (0.04748)	0.17243* (0.04749)
REG664	0.09689 (0.05374)	0.09498 (0.05346)	0.09008 (0.05314)	0.09057 (0.05314)	0.08755 (0.05313)	0.08671 (0.05317)
REG665	0.17920* (0.05610)	0.18129* (0.05582)	0.18016* (0.05539)	0.18092* (0.05540)	0.17996* (0.05536)	0.17941* (0.05539)
REG666	0.20056* (0.06236)	0.20398* (0.06204)	0.19998* (0.06156)	0.20099* (0.06158)	0.20017* (0.06153)	0.19953* (0.06156)
REG667	0.13442* (0.05986)	0.13458* (0.05955)	0.14092* (0.05911)	0.14152* (0.05912)	0.14026* (0.05908)	0.13973* (0.05910)
REG668						
REG669	0.17696* (0.05157)	0.17793* (0.05130)	0.18152* (0.05094)	0.18135* (0.05094)	0.18021* (0.05091)	0.18006* (0.05093)
R^2	0.2410	0.2492	0.2617	0.2620	0.2629	0.2630
Adjusted R^2	0.2358	0.2437	0.2555	0.2554	0.2563	0.2560
Observations	2039	2039	2039	2039	2039	2039

Works Cited

Card, D. (1993). Using Geographic Variation in College Proximity to Estimate the Return to Schooling. *NBER Working Paper Series*, 4483. <https://doi.org/10.3386/w4483>.